

ReadMe for the MassDEP PFAS TK Model

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Goals

This manual is meant to assist in running the population PFAS TK model developed by the Office of Research and Standards at the Massachusetts Department of Environmental Protection. This document is not intended to stand alone – please refer to the manuscript for the purpose and applications of the model.

Basic Instructions

Extract the zipped file PFAS_TK_Model to a relevant directory.

If you are testing the code and wish for it to run more quickly, you may wish to decrease the number of iterations in each exposure scenario. These can be changed in the inputdata Excel files in the Scenario tab under “niter”. I strongly recommend using the same “niter” for all exposure scenarios in a single inputdata file. Otherwise, some of the automatic post-processing will not work correctly.

Open the project “PFAS_TK_Model_MassDEP_V2.4.Rproj” in RStudio.

Open the first working script “2025-09-19_Set56_NewMatLISSMomF.R”

Run the functions starting from the top. The package.check process should facilitate installing any packages you don’t already have. The correct setID string is already entered.

The longest process to run is the function Model.All.Scenarios(sourceData), which may take 24 hours if using the original 35,000 iterations. It may run substantially faster on a more powerful computer.

After running all functions in “2025-09-19_Set56_NewMatLISSMomF.R”, check that the following files have been produced in the Set56_NewMatLISSMomF directory:

- Six files called RawData_Set56_NewMatLISSMomF_PFAS_F.RData, with each chemical replacing “PFAS” in the file name.
- SerumatDates_Set56_NewMatLISSMomF.RData

- LogsplineFits_Set56_NewMatLISSMomF.RData
- ProjPlotQuants_Set56_NewMatLISSMomF.RData
- ProjPlotQuants_Set56_NewMatLISSMomF.csv
- StabilityCheck_Set56_NewMatLISSMomF.csv
- ProjectionOutput_Set56_NewMatLISSMomF.csv

Restart your RStudio session, then re-open the project “PFAS_TK_Model.Rproj”. Open the second working script “2025-09-20_Set57_NewMatLIMdlMomM.R”. The correct setID and maternalID are already entered.

Run the functions starting from the top. Depending on your number of iterations, Model.All.Scenarios(sourceData) may take many hours.

After running all functions in “2025-09-20_Set57_NewMatLIMdlMomM.R”, check that the following files have been produced in the Set57_NewMatLIMdlMomM directory:

- Six files called RawData_Set57_NewMatLIMdlMomM_PFAS_M.RData, with each chemical replacing “PFAS” in the file name.
- SerumatDates_Set57_NewMatLIMdlMomM.RData
- ProjPlotQuants_Set57_NewMatLIMdlMomM.RData
- AllDatesQuants_Set57_NewMatLIMdlMomM.RData
- StabilityCheck_Set57_NewMatLIMdlMomM.csv
- ProjectionOutput_Set57_NewMatLIMdlMomM.csv
- ProjPlotQuants_Set57_NewMatLIMdlMomM.csv
- AllDatesQuants_Set57_NewMatLIMdlMomM.csv

Restart your RStudio session, then re-open the project “PFAS_TK_Model.Rproj”. Open the third working script “2025-09-24_Set58_NewSSMomEval.R”. The correct setID is already entered.

Run the functions starting from the top. Depending on your number of iterations, Model.All.Scenarios(sourceData) may take many hours.

After running all functions in “2025-09-24_Set58_NewSSMomEval.R”, check that the following files have been produced in the Set58_NewSSMomEval directory:

- Eleven files called RawData_Set58_NewSSMomEval_StudyLocation.RData, with each study location and demographic group replacing “StudyLocation” in the file name.
- SerumatDates_Set58_NewSSMomEval.RData
- EvalTable_Set58_NewSSMomEval.RData

- StabilityCheck_Set57_NewMatLIMdlMomM.csv
- EvalTable_Set58_NewSSMomEval.csv
- RMSEtable_Set58_NewSSMomEval.csv

General Process

Fundamentally, the TK model uses information about drinking water exposure scenarios to calculate distributions of serum PFAS for each scenario. Serum PFAS distributions are calculated at regular intervals over time as the population ages. The calculation is performed when parameters change, which is typically monthly below the age of 1 and annually beyond this age. The model tracks time by date, in order to reference exposure events.

The model main directory, called PFAS_TK_Model_MassDEP_V2.4, contains an R project, the working scripts for each model run, a “scripts” directory for the underlying functions, and directories for model runs. Each model run is called a “Set”, with a specified ID, e.g. “Set56_NewMatLISSMomF”. This setID must be the name of a folder in the R project directory. The setID will generally be appended to model-related files inside that directory. Within each set, several “exposure scenarios” can be modeled and analyzed, either individually or in relation to one another.

The inputs and analyses for each set are organized into a spreadsheet called “inputdata_setID”. Different tabs manage different parts of the modeling and analysis. The role of each tab is described in the “ReadMe” tab, along with the purpose of each column or row in the spreadsheet.

Functions in the “scripts” directory are generally well-commented, but I will give a brief overview here. In the order of their use, the functions are:

`Read.Input.Data()`: Reads and organizes the Excel file to be compatible with subsequent functions.

`Model.All.Scenarios()`: Organizes the rows of the inputdata file, each of which represents an exposure scenario, and runs them through the TK model process. Each exposure scenario has an ID to organize it. The process saves when each exposure scenario is complete. If the process is interrupted, re-running `Model.All.Scenarios()` will continue from exposure

scenarios without an associated output. However, if an exposure scenario needs to be re-run, delete the corresponding RawData file or list item in the RawData file. Otherwise the function will not replace the existing object.

`monte_carlo_vars()`: Draws parameters from the defined distributions to prepare for the modeling process. It creates a dataframe where each row represents a Monte Carlo iteration, and each column a parameter (or set of parameters) that will be used in the TK model. It uses a helper function `make_rnd()`, which reads the parameter entries and distribution shape to create random values from the specified distribution.

`calc_Serum_PFAS()`: The wrapper function for calculating a single iteration of serum changing over time. Produces a list of dataframes. Each list index represents an iteration. Each dataframe has rows representing dates and columns representing values such as consumed liquid concentration, background serum PFAS, and total serum PFAS at each date. This list is saved as the “results” object in the “RawData” files.

`create_frame()`: Sets up the dates for measuring serum and changing parameters. Before age 1, dates default to once per month aligned with the receptors’ birthday. Dates default to the receptors’ birthdays after age 1. Dates also include any time the PFAS concentration in water changes during the receptor’s lifetime. The time zoom functionality can generate extra dates to consider around a timeframe of interest. The function also generates a rounded-down age for eventual matching to liquid intakes. Until age 1, it rounds down to the nearest month, then divides by 12. Afterwards, it rounds down to the nearest year.

`pop_Liquid_Intake()`: Fills in the liquid intake in mL/(kg*day) at each date, multiplying by the liquid intake adjustment factor for the appropriate age group.

`pop_Liquid_Conc()`: Fills in water or breast milk PFAS concentrations as appropriate by date.

Water is filled in with a simple matching based on dates. Dates where receptor consumes breast milk instead of water are identified using `fed_mo`. The PFAS concentration in the milk is calculated using `MilkDecline`, `LactXfer`, and `MatStart` (appearing in the function as `momserum`).

`pop_Bkgd_Serum()`: Fills in background serum PFAS for each date. It also makes the background PFAS phase in linearly starting from 0 until age 3.

`calc_Total_Serum()`: Uses entries in each date to calculate `Total_Serum` at the next date. The receptor begins with `Total_Serum` based on placental transfer, with no background. The second row needs to be made manually because there isn’t a value for `bkgdold` before birth

(so all the placental transfer PFAS is available for elimination). Uses the helper function `nextserum()` to perform the calculation using the integrated rate law for the system.

`Extract.Serum.at.Dates()`: Combines all iterations' serum values into a single dataframe, where rows are iterations and columns are scenario IDs with dates. Loads each RawData file one by one, to avoid overwhelming small computers. May take several hours if using 35,000 iterations.

Downstream analyses are relatively self-explanatory.

`Fit.Serum.Distributions()`: For when model output will be used as the maternal serum distribution in another exposure scenario.

`Evaluate.Against.Observed.Sera()`: For comparing modeled serum percentiles against observed serum percentiles.

`Project.from.Serum()`: For calculating CGWCs and other linear regressions. Connects exposure scenario parameters (for example, water concentration) to serum concentration at a specified percentile.

`Prep.Projection.Plot.Data()`: Saves data in the format required to plot the CGWC projection process.

`Prep.Lifetime.Plot.Data()`: Saves data in the format required to plot serum percentiles over the lifetime of a population.

`Check.Stability()`: A process for estimating Monte Carlo error based on principles of quantile regression. Helps determine if enough iterations were run.

`Calculate.RMSE()`: Calculates RMSE between observed and modeled percentiles.